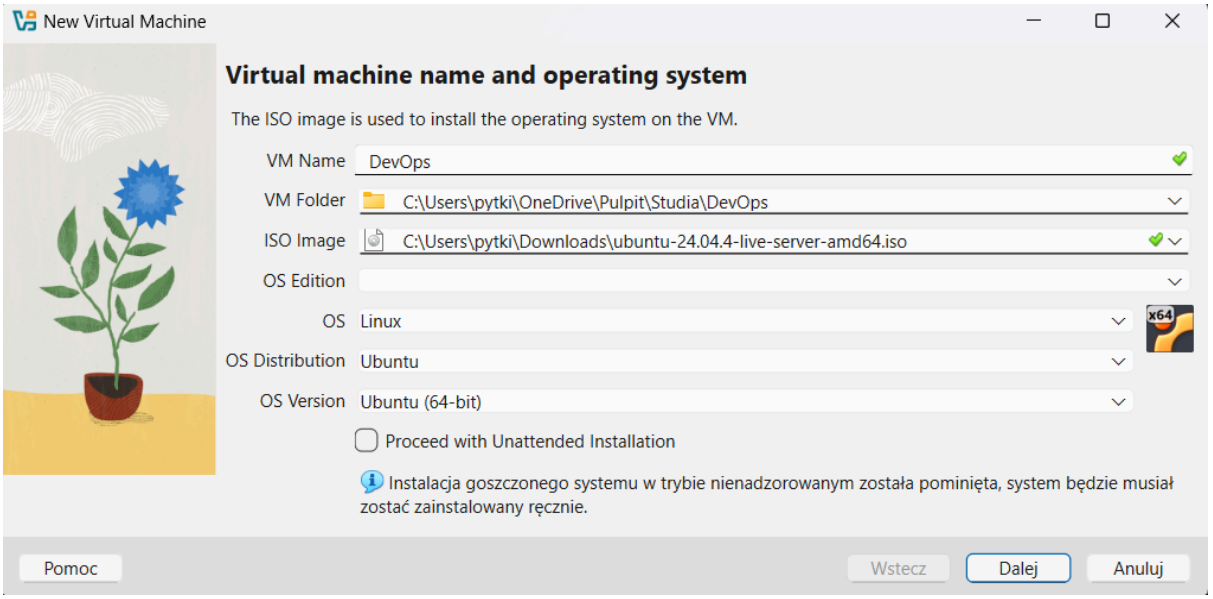
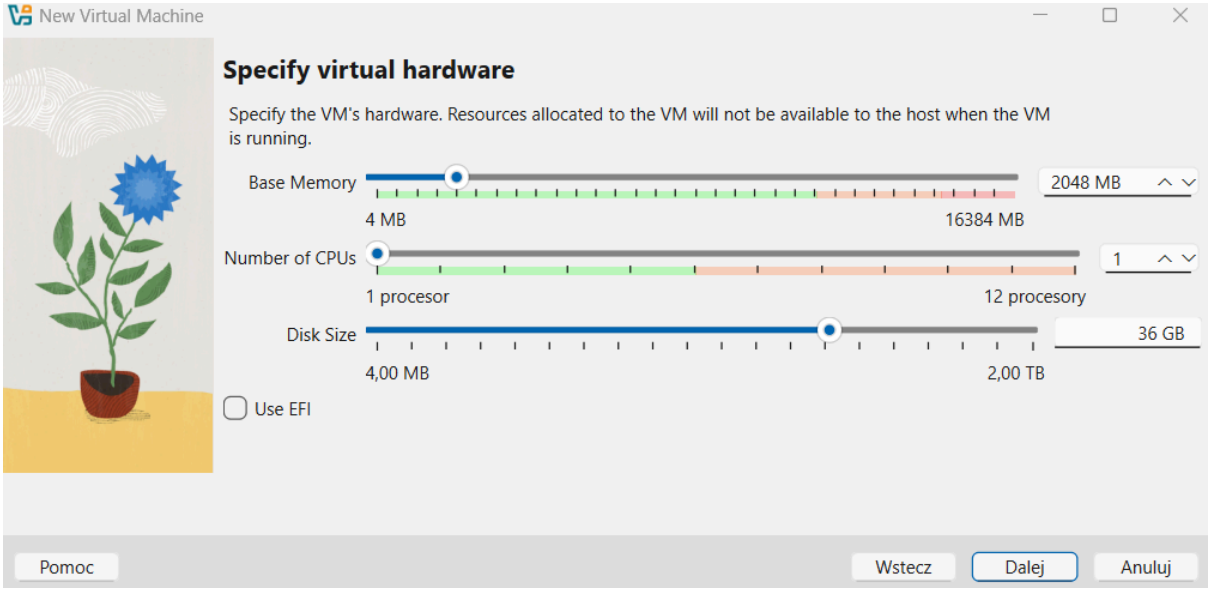
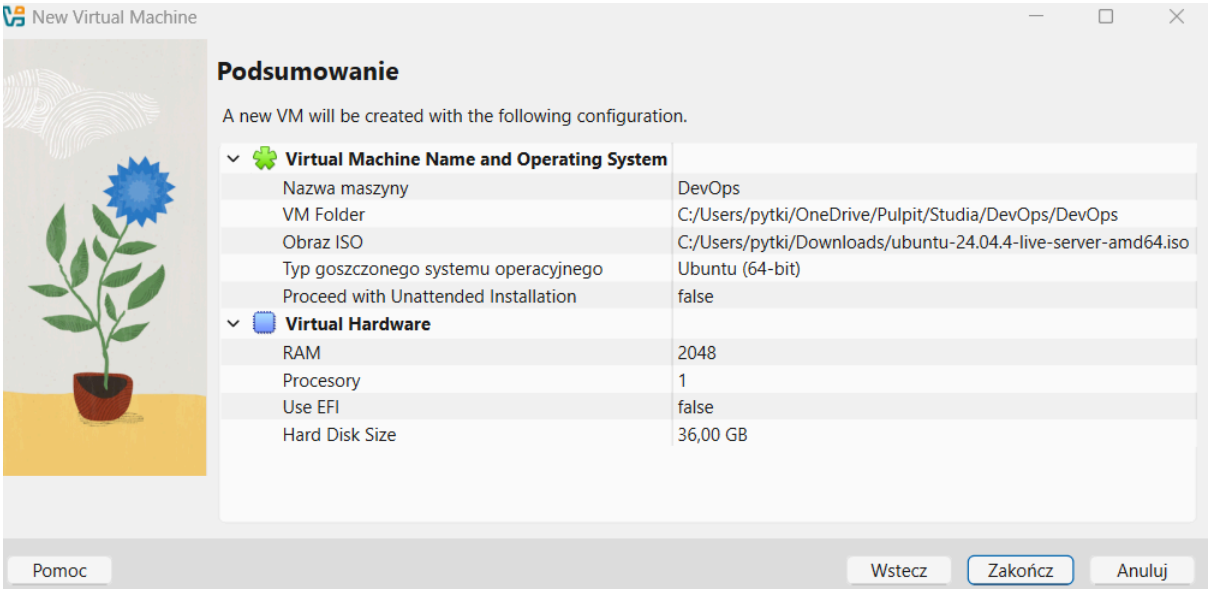
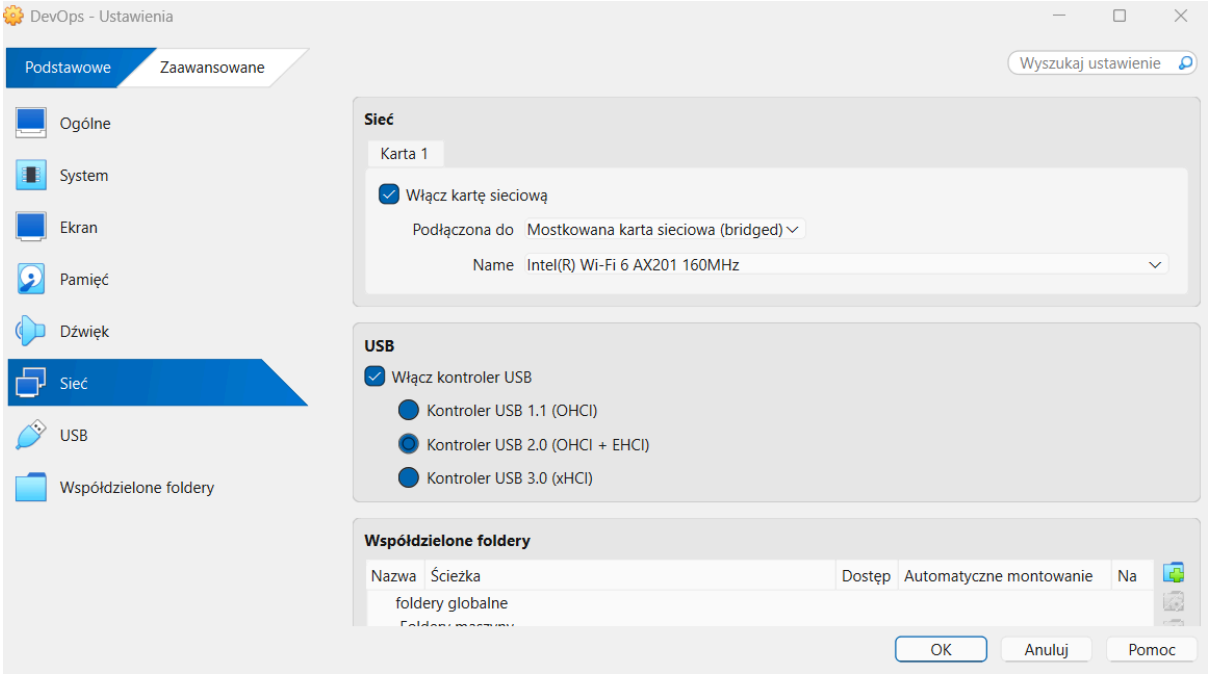
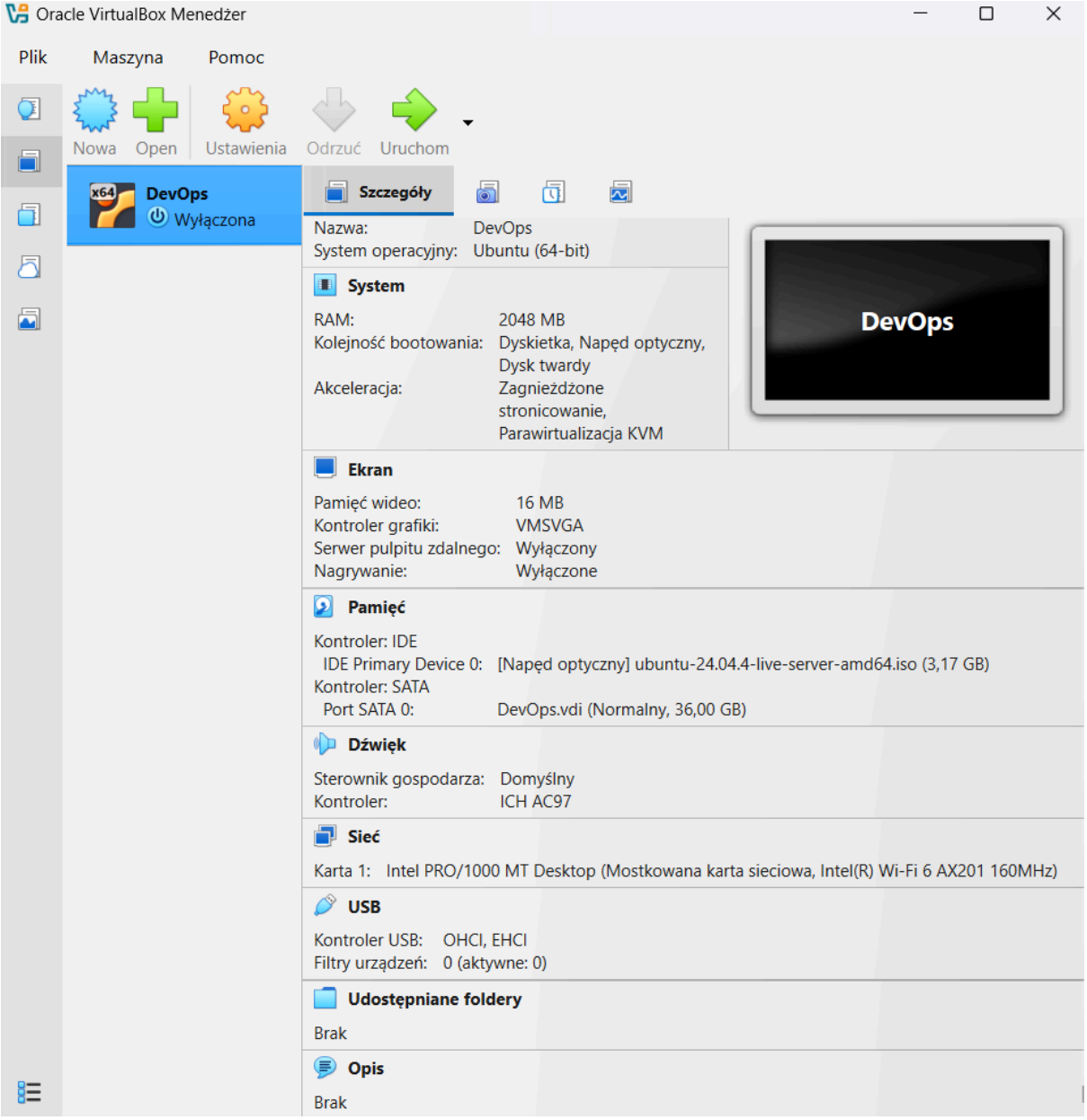
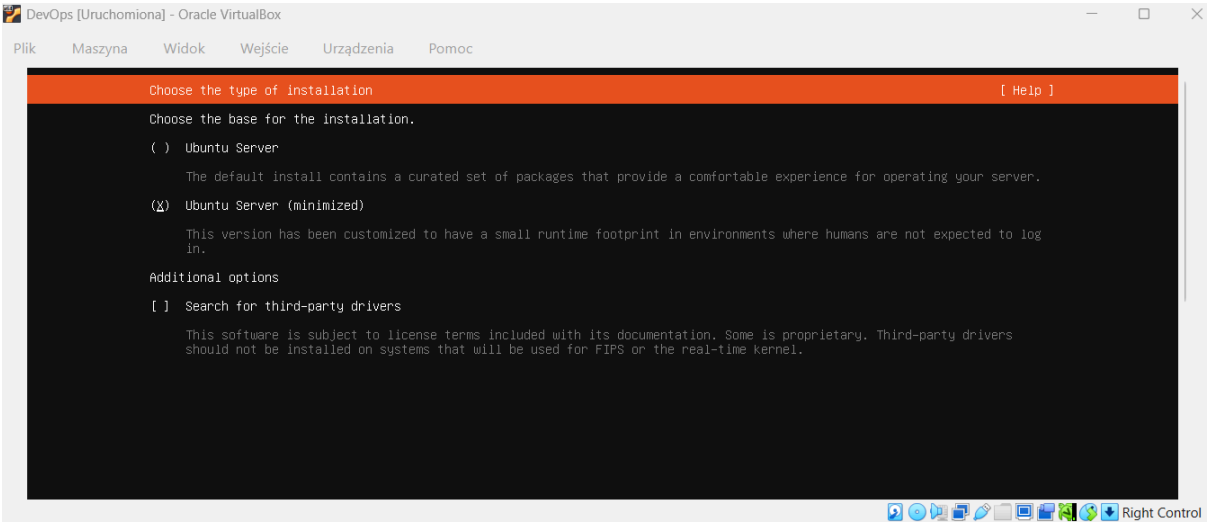

New Virtual Machine
Virtual machine name and operating system
The ISO image is used to install the operating system on the VM.
VM Name
DevOps
VM Folder
C:\Users\pytki\OneDrive\Pulpit\Studia\DevOps
ISO Image
C:\Users\pytki\Downloads\ubuntu-24.04.4-live-server-amd64.iso
OS Edition
OS
Linux
OS Distribution
Ubuntu
OS Version
Ubuntu (64-bit)
Proceed with Unattended Installation
Instalacja goszczonego systemu w trybie nienadzorowanym została pominięta, system będzie musiał zostać zainstalowany ręcznie.
Pomoc
Wstecz
Dalej
Anuluj


New Virtual Machine
Specify virtual hardware
Specify the VM's hardware. Resources allocated to the VM will not be available to the host when the VM is running.
Base Memory
2048 MB
4 MB
16384 MB
Number of CPUs
1
1 procesor
12 procesory
Disk Size
36 GB
4,00 MB
2,00 TB
Use EFI
Pomoc
Wstecz
Dalej
Anuluj


New Virtual Machine
Podsumowanie
A new VM will be created with the following configuration.
Virtual Machine Name and Operating System
Nazwa maszyny
DevOps
VM Folder
C:/Users/pytki/OneDrive/Pulpit/Studia/DevOps/DevOps
Obraz ISO
C:/Users/pytki/Downloads/ubuntu-24.04.4-live-server-amd64.iso
Typ goszczonego systemu operacyjnego
Ubuntu (64-bit)
Proceed with Unattended Installation
false
Virtual Hardware
RAM
2048
Procesory
1
Use EFI
false
Hard Disk Size
36,00 GB
Pomoc
Wstecz
Zakończ
Anuluj


DevOps - Ustawienia
Podstawowe
Zaawansowane
Wyszukaj ustawienie
Ogólne
System
Ekran
Pamięć
Dźwięk
Sieć
USB
Współdzielone foldery
Sieć
Karta 1
Włącz kartę sieciową
Podłączona do
Mostkowana karta sieciowa (bridged)
Name
Intel(R) Wi-Fi 6 AX201 160MHz
USB
Włącz kontroler USB
Kontroler USB 1.1 (OHCI)
Kontroler USB 2.0 (OHCI + EHCI)
Kontroler USB 3.0 (xHCI)
Współdzielone foldery
Nazwa
Ścieżka
Dostęp
Automatyczne montowanie
Na
foldery globalne
OK
Anuluj
Pomoc


Oracle VirtualBox Menedżer
Plik
Maszyna
Pomoc
Nowa
Open
Ustawienia
Odrzuć
Uruchom
DevOps
Wyłączona
Szczegóły
Nazwa: DevOps
System operacyjny: Ubuntu (64-bit)
System
RAM: 2048 MB
Kolejność bootowania: Dyskietka, Napęd optyczny, Dysk twardy
Akceleracja: Zagnieżdżone stronicowanie, Parawirtualizacja KVM
DevOps
Ekran
Pamięć wideo: 16 MB
Kontroler grafiki: VMSVGA
Serwer pulpitu zdalnego: Wyłączony
Nagrywanie: Wyłączone
Pamięć
Kontroler: IDE
IDE Primary Device 0: [Napęd optyczny] ubuntu-24.04.4-live-server-amd64.iso (3,17 GB)
Kontroler: SATA
Port SATA 0: DevOps.vdi (Normalny, 36,00 GB)
Dźwięk
Sterownik gospodarza: Domyślny
Kontroler: ICH AC97
Sieć
Karta 1: Intel PRO/1000 MT Desktop (Mostkowana karta sieciowa, Intel(R) Wi-Fi 6 AX201 160MHz)
USB
Kontroler USB: OHCI, EHCI
Filtry urządzeń: 0 (aktywne: 0)
Udostępniane foldery
Brak
Opis
Brak


DevOps [Uruchomiona] - Oracle VirtualBox
Plik
Maszyna
Widok
Wejście
Urządzenia
Pomoc
Choose the type of installation
[Help]
Choose the base for the installation.
( ) Ubuntu Server
The default install contains a curated set of packages that provide a comfortable experience for operating your server.
(X) Ubuntu Server (minimized)
This version has been customized to have a small runtime footprint in environments where humans are not expected to log in.
Additional options
[ ] Search for third-party drivers
This software is subject to license terms included with its documentation. Some is proprietary. Third-party drivers should not be installed on systems that will be used for FIPS or the real-time kernel.
Right Control

DevOps [Uruchomiona] - Oracle VirtualBox

Plik Maszyna Widok Wejście Urządzenia Pomoc

```
Network configuration                                                    [ Help ]

Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient
access for updates.

  NAME    TYPE  NOTES
[ enp0s3  eth   -                ▸ ]
  DHCPv4  192.168.1.33/24
  08:00:27:22:86:cf / Intel Corporation / 82540EM Gigabit Ethernet Controller (PRO/1000 MT Desktop Adapter)

[ Create bond ▸ ]
```

```
Storage configuration

FILE SYSTEM SUMMARY

  MOUNT POINT     SIZE     TYPE      DEVICE TYPE
[ /              35.997G  new ext4  new partition of local disk ▸ ]


AVAILABLE DEVICES

  No available devices

[ Create software RAID (md) ▸ ]
[ Create volume group (LVM) ▸ ]


USED DEVICES

  DEVICE                                          TYPE           SIZE
[ VBOX_HARDDISK_VB3c01ea14-17a8ee08               local disk    36.000G  ▸ ]
  partition 1  new, BIOS grub spacer                               1.000M  ▸
  partition 2  new, to be formatted as ext4, mounted at /         35.997G  ▸
```

```
Ubuntu 24.04.4 LTS devops tty1

devops login: karro
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

karro@devops:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ce:c8:a3 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.34/24 metric 100 brd 192.168.1.255 scope global dynamic enp0s3
       valid_lft 86390sec preferred_lft 86390sec
    inet6 fe80::a00:27ff:fece:c8a3/64 scope link
       valid_lft forever preferred_lft forever
karro@devops:~$ _
```

```
PS C:\Users\pytki> ssh karro@192.168.1.34
The authenticity of host '192.168.1.34 (192.168.1.34)' can't be established.
ED25519 key fingerprint is SHA256:ET9Qyc+89+x1XodouORyEgXuWVGQZhVZ/y6xzyMcqkg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.34' (ED25519) to the list of known hosts.
karro@192.168.1.34's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

karro@devops:~$
```

```
karro@devops:~$ sudo apt install git-all
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils cvs cvsps git git-cvs git-doc git-email git-gui git-man git-mediawiki git-svn
  gitk gitweb less libalgorithm-c3-perl libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 libauthen-sasl-perl
  libb-hooks-endofscope-perl libb-hooks-op-check-perl libcgi-fast-perl libcgi-pm-perl libclass-c3-perl libclass-c3-xs-perl
  libclass-data-inheritable-perl libclass-inspector-perl libclass-method-modifiers-perl libclass-singleton-perl
  libclass-xsaccessor-perl libclone-perl libcommon-sense-perl libdata-dump-perl libdata-optlist-perl
  libdatetime-format-builder-perl libdatetime-format-iso8601-perl libdatetime-format-strptime-perl libdatetime-locale-perl
  libdatetime-perl libdatetime-timezone-perl libdbd-sqlite3-perl libdbi-perl libdevel-callchecker-perl libdevel-caller-perl
  libdevel-lexalias-perl libdevel-stacktrace-perl libdigest-bubblebabble-perl libdigest-hmac-perl libdynaloader-functions-perl
  libemail-valid-perl libencode-locale-perl liberror-perl libeval-closure-perl libexception-class-perl libfcgi-bin libfcgi-perl
  libfcgi0t64 libfile-listing-perl libfile-sharedir-perl libfont-afm-perl libhtml-form-perl libhtml-format-perl
  libhtml-parser-perl libhtml-tagset-perl libhtml-tree-perl libhttp-cookies-perl libhttp-daemon-perl libhttp-date-perl
  libhttp-message-perl libhttp-negotiate-perl libidn12 libio-html-perl libio-socket-ssl-perl libjansson4 libjson-perl
  libjson-xs-perl liblua5.4-0 liblwp-mediatypes-perl liblwp-protocol-https-perl libmailtools-perl libmediawiki-api-perl
  libmodule-implementation-perl libmodule-runtime-perl libmro-compat-perl libnamespace-autoclean-perl libnamespace-clean-perl
  libnet-dns-perl libnet-dns-sec-perl libnet-domain-tld-perl libnet-http-perl libnet-libidn-perl libnet-smtp-ssl-perl
  libnet-ssleay-perl libpackage-stash-perl libpackage-stash-xs-perl libpadwalker-perl libparams-classify-perl
  libparams-util-perl libparams-validate-perl libparams-validationcompiler-perl libperl4-corelibs-perl libreadonly-perl
  libref-util-perl libref-util-xs-perl librole-tiny-perl libserf-1-1 libspecio-perl libsub-exporter-perl
  libsub-exporter-progressive-perl libsub-identify-perl libsub-install-perl libsub-name-perl libsub-quote-perl libsvn-perl
  libsvn1 libtcl8.6 libtimedate-perl libtk8.6 libtry-tiny-perl libtypes-serialiser-perl liburi-perl libutf8proc3
  libvariable-magic-perl libwww-perl libwww-robotrules-perl libxft2 libxrender1 libxss1 libxstring-perl libyaml-libyaml-perl
  libyaml-perl patch perl-openssl-defaults ssl-cert tcl tcl8.6 tk tk8.6 x11-common
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser ufw mksh rcs git-daemon-run | git-daemon-sysvinit
  aspell meld mediawiki subversion libgssapi-perl libmldbm-perl libnet-daemon-perl libsql-statement-perl
  libio-compress-brotli-perl libcrypt-ssleay-perl libscalar-number-perl libtest-fatal-perl libbusiness-isbn-perl
```

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to authenticate to the API over Basic Authentication.

Note

DevOps

What's this token for?

Expiration **Select date ***

Custom 01.07.2026

The token will expire on the selected date

Select scopes

Scopes define the access for personal tokens. Read more about OAuth scopes.

☑ **repo**	Full control of private repositories
☑ repo:status	Access commit status
☑ repo_deployment	Access deployment status
☑ public_repo	Access public repositories
☑ repo:invite	Access repository invitations
☑ security_events	Read and write security events
☑ **workflow**	Update GitHub Action workflows

Tokens you have generated that can be used to access the GitHub API.

DevOps — *admin:public_key, admin:ssh_signing_key, repo, workflow* Never used Delete

Expires ***on Wed, Jul 1 2026.***

```
karro@devops:~$ git --version
git version 2.43.0
karro@devops:~$
```

```
karro@devops:~$ git clone https://github.com/InzynieriaOprogramowaniaAGH/MDO2026_ITE
Cloning into 'MDO2026_ITE'...
remote: Enumerating objects: 130, done.
remote: Counting objects: 100% (98/98), done.
remote: Compressing objects: 100% (83/83), done.
remote: Total 130 (delta 8), reused 98 (delta 8), pack-reused 32 (from 1)
Receiving objects: 100% (130/130), 1.75 MiB | 30.00 KiB/s, done.
Resolving deltas: 100% (13/13), done.
karro@devops:~$
```

```
karro@devops:~$ ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/karro/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/karro/.ssh/id_ed25519
Your public key has been saved in /home/karro/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:txsM6Flj8FWDemn4NKXrsLb0mbJYgjnJY6bICNcDEqg karro@devops
The key's randomart image is:
+--[ED25519 256]--+
|           .o    |
|.         ....   |
|o     .  o.+     |
|..     +o.B      |
|E .   . S=.o     |
| . + = +.=o.     |
|. . % + o++      |
|+o + + =+..=     |
|o..   ..o+=      |
+----[SHA256]-----+
karro@devops:~$
```

```
karro@devops:~$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAINKKr17To7zNDNmowE9EwLEAIFFiiO/PDS+ypHbre5Nw karro@devops
karro@devops:~$
```

Add new SSH Key

Title

DevOps

Key type

Authentication Key

Key

AAAAC3NzaC1lZDI1NTE5AAAAINKKr17To7zNDNmowE9EwLEAIFFiiO/PDS+ypHbre5Nw

Add new SSH Key

Title

DevOps

Key type

Authentication Key

Key

ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAINKKr17To7zNDNmowE9EwLEAIFFiiO/PDS+ypHbre5Nw karro@devops

Add SSH key

SSH keys

New SSH key

This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.

Authentication keys

SSH

DevOps

SHA256:txsM6Flj8FWDemn4NKXrsLb0mbJYgjnJY6bICNcDEqg

Added on Mar 4, 2026

Never used — Read/write

Delete

Check out our guide to connecting to GitHub using SSH keys or troubleshoot common SSH problems.

```
karro@devops:~$ git clone git@github.com:InzynieriaOprogramowaniaAGH/MDO2026_ITE.
Cloning into 'MDO2026_ITE.com'...
The authenticity of host 'github.com (140.82.121.4)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
```

```
karro@devops:~/devops$ git clone git@github.com:InzynieriaOprogramowaniaAGH/MDO2026_ITE.git
Cloning into 'MDO2026_ITE'...
remote: Enumerating objects: 130, done.
remote: Counting objects: 100% (98/98), done.
remote: Compressing objects: 100% (83/83), done.
remote: Total 130 (delta 8), reused 98 (delta 8), pack-reused 32 (from 1)
Receiving objects: 100% (130/130), 1.75 MiB | 24.00 KiB/s, done.
Resolving deltas: 100% (13/13), done.
```

sftp://karro@192.168.1.34 - FileZilla

File Edit View Transfer Server Bookmarks Help

Host: tp://192.168.1.34 Username: karro Password: •••••••• Port: Quickconnect

```
tatus:    Connecting to 192.168.1.34...
tatus:    Using username "karro".
tatus:    Connected to 192.168.1.34
tatus:    Retrieving directory listing...
tatus:    Listing directory /home/karro
tatus:    Directory listing of "/home/karro" successful
```

sftp://karro@192.168.1.34 - FileZilla

File Edit View Transfer Server Bookmarks Help

Host: tp://192.168.1.34 Username: karro Password: •••••••• Port: Quickconnect

```
Status:   Connecting to 192.168.1.34...
Status:   Using username "karro".
Status:   Connected to 192.168.1.34
Status:   Retrieving directory listing...
Status:   Listing directory /home/karro
Status:   Directory listing of "/home/karro" successful
Status:   Connecting to 192.168.1.34...
Status:   Using username "karro".
Status:   Connected to 192.168.1.34
Status:   Starting upload of C:\Users\pytki\OneDrive\Pulpit\Test.txt
Status:   File transfer successful, transferred 16 bytes in 1 second
Status:   Retrieving directory listing of "/home/karro"...
Status:   Listing directory /home/karro
Status:   Directory listing of "/home/karro" successful
```

Remote site: /home/karro

- /
  - home
    - karro
      - .cache
      - .dotnet
      - .ssh

Filename	Filesize	Filetype	Last modifi...	Permissi...	Owner/Gr...
.dotnet		Folder pl...	04.03.2026 ...	drwxrwx...	karro karro
.ssh		Folder pl...	04.03.2026 ...	drwx------	karro karro
.vscode-server		Folder pl...	04.03.2026 ...	drwxr-x---	karro karro
devops		Folder pl...	04.03.2026 ...	drwxrwx...	karro karro
MDO2026_ITE		Folder pl...	04.03.2026 ...	drwxrwx...	karro karro
.bash_history	5	Plik BAS...	04.03.2026 ...	-rw-------	karro karro
.bash_logout	220	Bash Lo...	31.03.2024 ...	-rw-r--r--	karro karro
.bashrc	3 771	Bash RC ...	31.03.2024 ...	-rw-r--r--	karro karro
.profile	807	Profile S...	31.03.2024 ...	-rw-r--r--	karro karro
.sudo_as_admin_...	0	Plik SUD...	04.03.2026 ...	-rw-r--r--	karro karro
.wget-hsts	183	Plik WGE...	04.03.2026 ...	-rw-rw-r--	karro karro
Test.txt	16	Dokume...	04.03.2026 ...	-rw-rw-r--	karro karro

7 files and 6 directories. Total size: 5 002 bytes

Setting up SSH Host 192.168.1.34: Downloading VS Code Server

SSH: 192.168.1.34

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
karro@devops:~$ cd cd ~/devops/MDO2026_ITE
bash: cd: too many arguments
karro@devops:~$ cd ~/devops/MDO2026_ITE
karro@devops:~/devops/MDO2026_ITE$ mkdir KP419785
karro@devops:~/devops/MDO2026_ITE$ echo "tekst" > KP419785/text.txt
```

```
karro@devops:~/devops/MDO2026_ITE$ git checkout -b KP419785 grupa5
A       KP419785/text.txt
Switched to a new branch 'KP419785'
karro@devops:~/devops/MDO2026_ITE$
```

```
karro@devops:~/devops/MDO2026_ITE$ git branch
* KP419785
  grupa5
  main
```

```
karro@devops:~/devops/MDO2026_ITE$ git add .
```

```
karro@devops:~/devops/MDO2026_ITE$ git commit -m "branch created"
[KP419785 388abdd] branch created
 1 file changed, 1 insertion(+)
 create mode 100644 KP419785/text.txt
karro@devops:~/devops/MDO2026_ITE$
```

```
karro@devops:~/devops/MDO2026_ITE$ git push
fatal: The current branch KP419785 has no upstream branch.
To push the current branch and set the remote as upstream, use

    git push --set-upstream origin KP419785

To have this happen automatically for branches without a tracking
upstream, see 'push.autoSetupRemote' in 'git help config'.

karro@devops:~/devops/MDO2026_ITE$  git push --set-upstream origin KP419785
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 390 bytes | 390.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'KP419785' on GitHub by visiting:
remote:      https://github.com/InzynieriaOprogramowaniaAGH/MDO2026_ITE/pull/new/KP419785
remote:
To github.com:InzynieriaOprogramowaniaAGH/MDO2026_ITE.git
 * [new branch]      KP419785 -> KP419785
branch 'KP419785' set up to track 'origin/KP419785'.
karro@devops:~/devops/MDO2026_ITE$
```